Major Project-1
End Semester Report (2023)
For
"Text to Insight: Summarise & Respond (TSR)"

Submitted By:

Specialization	SAP ID	Name
B.Tech-CSE(AIML)	500086120	SAKSHAM RATAN
B.Tech-CSE (AIML)	500086138	SIDDHARTH SINGH
B.Tech-CSE (AIML)	500084888	TUSHAR KALRA
B.Tech-CSE (AIML)	500085007	VEER SINGH RYAIT



Cluster - AIML

School of Computer Science

UNIVERSITY OF PETROLEUM & ENERGY STUDIES,

DEHRADUN- 248007, Uttarakhand

Dr. Sanjeev Kumar
Project Guide

Dr. Anil Kumar
Cluster Head

1. Abstract

The proposed major project delves into the realm of Document Interpretation, a critical component of information retrieval and knowledge management. This comprehensive endeavor encompasses two key components: Text Summarization and Question Answering on documents.The first component of the project deals with text summarization. In the approach used the documents are first divided into small segments called chunks . Each of the segments is then summarized. In the end all the summaries are joined to form the final summary The second component is question answering . In this the user can query the document and find the most contextually relevant text from the document .This will save the user a lot of time. When presented with a user's query, the system initiates a retrieval step to fetch relevant documents. These documents, along with the original question, are then fed into the language model. The model leverages this combined input to generate a precise and contextually relevant response. This project aims to utilize the power of LLMS (large language models) unlike traditional summary techniques which fail to give contextual response and give summary which does not encompass the entire information in the document.

Keywords— Large language Models, Text Summarization, Question Answering

2. Introduction

The task of extracting valuable insights from vast textual datasets has been a longstanding challenge. Various other approaches to text summarization such as rule based techniques or extractive techniques have been used, while the question-answering systems relied on structured data sources or manually curated knowledge bases. Therefore, the problem we aim to address in this project is to develop a solution that capitalizes on the capabilities of LLMs, thereby redefining the boundaries of what is achievable in document understanding by text summarization and querying over the given textual information.

The purpose of the project is to create a Text Summarization System for concise and informative summaries from lengthy text content while maintaining high accuracy. Build an Insightful Response Generation system capable of generating insightful responses to user queries. Evaluate the effectiveness and usability of the summarization and insight generation processes.

The prime beneficiaries of this project encompass a diverse range of stakeholders. Researchers and academics stand to gain a powerful tool for expediting the extraction of critical information from extensive textual resources. This will notably enhance their research efficiency and deepen their comprehension. Likewise, content creators and curators will find immense value in the system's ability to swiftly distill large volumes of text into concise, informative content suitable for various platforms. Information seekers, whether professionals or individuals, will benefit from the system's capacity to provide insightful responses to their queries, significantly streamlining their search process. Businesses and enterprises can leverage this tool for expeditious document analysis, facilitating the rapid extraction of relevant insights crucial for informed decision-making. Educational institutions, too, can employ this system as a valuable learning aid, aiding students in understanding and summarizing complex textual materials. Lastly, general users grappling with substantial volumes of text, such as articles, reports, or documents, will find this system invaluable for swiftly obtaining summarized content and

pertinent responses to their queries.
The project aims to develop a Text Summarization & Insightful Response (TSR) system utilizing advanced Large Language Models (LLMs). This system targets the enhancement of document understanding by generating concise, informative summaries from extensive text and providing insightful responses to user queries. The objectives include creating a robust summarization tool and a responsive QA system, ultimately improving information retrieval. The deliverables will consist of the fully functional TSR software, along with documentation and user guides. The project's scope extends to applications in fields such as content curation, research, and information extraction for improved efficiency and comprehension**.**

3. Problem Identification

The task of extracting valuable insights from vast textual datasets has been a longstanding challenge. Various other approaches to text summarization such as rule based techniques or extractive techniques have been used, while the question-answering systems relied on structured data sources or manually curated knowledge bases. Therefore, the problem we aim to address in this project is to develop a solution that capitalises on the capabilities of LLMs, thereby redefining the boundaries of what is achievable in document understanding by text summarization and querying over the given textual information.

4. Objectives

1. Create Text Summarization System for concise and informative summaries from lengthy text content while maintaining high accuracy.

2. Build an Insightful Response Generation system capable of generating insightful responses to user queries.

3. Evaluate the effectiveness and usability of the summarization and insight generation processes.

5. Literature Survey

A documents summarizer helps in analyzing the gist of the large text collection quickly and also saves time by avoiding reading the whole document[3]. Ingesting documents into a queryable format and coupling this with the retrieval augmented generation chain, we enable the model to fetch and incorporate relevant documents in real-time when posed with questions[1]. Handling large texts is a very difficult task. Methods such as stemming , text clustering , information extraction , storing and fetching unstructured data are used [2]. A well-liked programming model for processing large data sets is MapReduce. A modified K-means clustering algorithm based on MapReduce framework is proposed to perform clustering on large data sets[3]. It underscores the models' capacity to generalize and execute tasks with minimal training examples. The emphasizing and the versatility of pre-trained language models across various applications marks a significant advancement in natural language processing[1]. Evolution in NLP's and CI's call attention to the flexibility of language models, particularly as the task of making prognosis based on a tight amount of samples. Revolution in encapsulation techniques, clustering techniques, and coherent question-answering have been remarkable. There is also a

focus on knowledge-based visual question answering and text summarization with cognitive intelligence. The developing topography incorporates calls for a new training model for engine-based models and insights into observing prolific artificial intelligence models using measures like BLEU score.

In the domain of natural language processing, various studies have been conducted to study and improve on existing textual question answering(QA) models. A study conducted in 2018 aimed to compare two state of the art models: DCN+ and S-Reader on the SQuAD dataset and other variants of the dataset [4]. Another study presented the Prophet framework. This innovative approach aims to enhance the performance of knowledge-based visual question answering (VQA) by leveraging large language models like GPT-3. This technique aims towards enhancing the execution of VQA by using LLM's like GPT-3. Contrasted from previous techniques that gave limited input information to GPT-3, a vanilla VQA model is trained to extract response information, using GPT-3 with more specific information. This technique overcomes existing models on stimulating knowledge based VQA records, OK-VQA and A-OKVQA [5].

6. Existing System

In today's world, text summarization takes use of two main methods, i.e, extracting important sentences from the original source and creating a smaller version by being creative. Also, for insightful responses, the machine aims to not only just understand what the user is trying to convey, but also clever responses. Generating a concise and precise summary of voluminous texts is the key idea in summarization[6]. Still, there are obstacles like handling a variety of data and making everyone happy. Developers around the world are working to make these systems better.

7. Proposed System Design

Our proposed system develops a Text Summarization and an Insightful Response Generation system to increase the efficiency in handling lengthy text and queries. The main focus is on generating short and clear informative summaries with high accuracy, as well as generating insightful responses to the queries.

8. Methodology

Our Text Summarization and Insightful Response project relies on the MapReduce framework and retrieval augmented generation. The process begins with data acquisition from a variety of sources, such as text documents, web pages.Text summarization is a process of grouping key points of the text from a large collection of text sources[7]. The collection of sources can be reviewed from the web, documents, blog posts, social media posts, news feeds, etc. Generating a concise and precise summary of voluminous texts is the key idea in summarization[7]. Subsequently, data preprocessing is performed to remove the irrelevant information, like punctuation and special characters, and to structure the text for compatibility with MapReduce. The Map phase follows, where data is transformed into key-value pairs, these keys represent specific attributes of the text, and the values contain corresponding text segments. Mappers handle tasks such as word counting, sentence segmentation, and feature extraction in preparation for text summarization. Following the Map phase, the Shuffle and Sort phase is initiated to

arrange the key-value pairs, grouping related data and organizing it for the Reduce phase. The Reduce phase is where the main work happens. Summarization methods may encompass extractive or abstractive approaches, involving sentence ranking and content selection. Insightful response generation leverages natural language processing techniques and context-awareness to create responses that are informative and contextually relevant from the ingestion of documents into a queryable format and then the retrieval augmented generation chain.

1. Text Summarization System, which calls for the creation of a better tool for accurately condensing large texts. Our objective is to create summaries from text that are helpful and maintain key information. This technique will split the document into small chunks that fit the model. It will begin by providing a summary of each section, and then it will compile a summary of the summaries. MapReduce uses two prompts: the first, the map function, to summarize each data chunk, and a second, the combine or reduce function, to combine each summary into the final one.

2. Generating insightful responses for our goal to build a responsive system that can respond to queries or questions from users by providing insights. Question answering involves fetching multiple documents, and then asking a question from them. The Large Language Model response will have the answer to your question, based on the content of the documents. The approach for the question answering is Similarity Search.

First we prepare the data and do a similarity search over a chroma vector database that runs on the local machine as a library. It is one of the good common ways for search and store over unstructured data and to embed it and to store the resulting embedding vectors, and at query time embedding the unstructured query and retrieve the embedding vectors which are 'most similar' to the embedded query and this showcases basic concept related to vector stores. A key part of working with vector stores is creating the vector to put in them using embeddings.

9. Evaluation Methodology

Metrics ROUGE Score: Evaluating the summaries using Recall-Oriented Understudy for Gisting Evaluation. ROUGE is a metric that measures the overlap between the generated text and the reference text in terms of recall[9][10].

$$\text{ROUGE} = \frac{\text{number of n-gram found in model and reference}}{\text{number of n-gram in reference}}$$

```
from rouge import Rouge
reference = 'this movie was awesome'
candidate = 'this movie was awesome too'
rouge = Rouge()
scores = rouge.get_scores(candidate, reference)[0]
['rouge-2']
['f']
print(scores)
```

```
'f': 0.8888888839506174
```

10. UML Diagram

1.Flow Chart

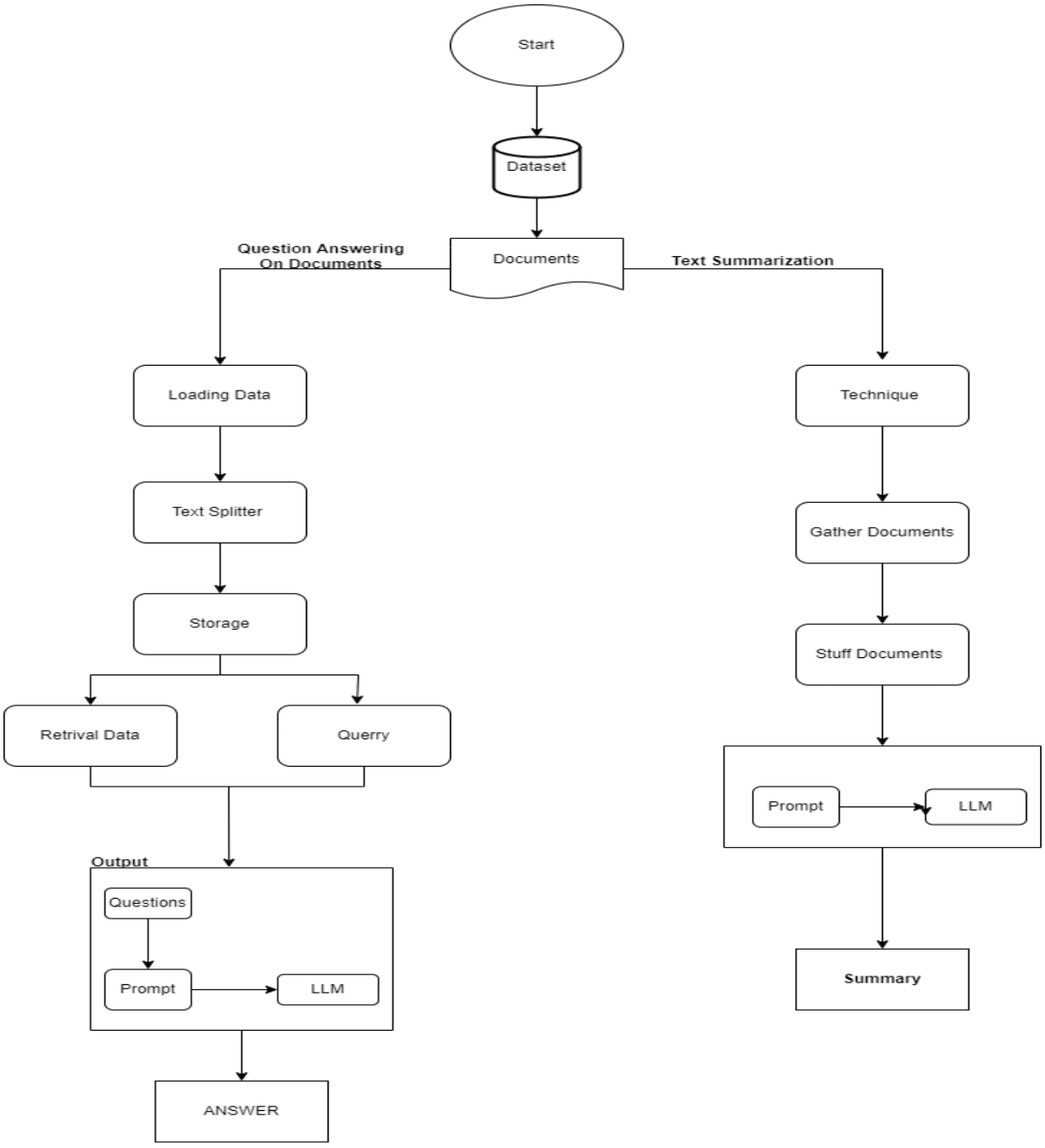


Fig 10.1: Flowchart for the system

2. Use Case Diagram:

According to Fig. 10.2, it expresses what are the main functions our system provides. Use Case Diagram for our system:

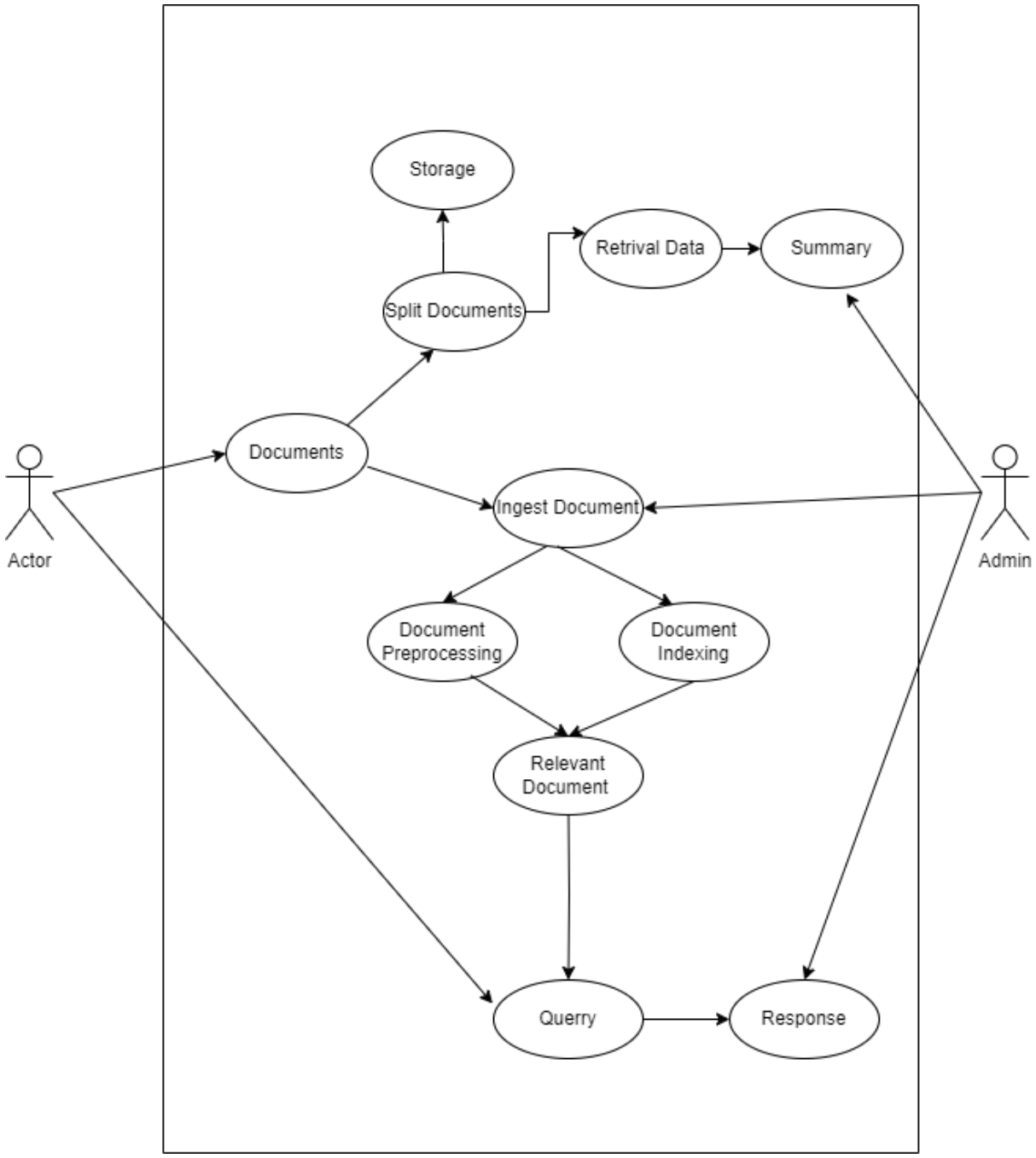


Figure 10.2: Use Case Diagram of the system

11. Results




```
os.environ['OPENAI_API_KEY'] = credentials.openai_api_key

map_prompt = """
Write a concise summary of the following in at least 250 words and at most 280 words:
"{text}"
CONCISE SUMMARY:
"""

map_prompt_template = PromptTemplate(template=map_prompt, input_variables=["text"])


combine_prompt = """
Write a concise summary of the following text delimited by triple backquotes. Consider each different summary to be a part of the same article.
Return your response in at least 500 words and at most 550 words which covers the key points of the text.
```{text}```
SUMMARY:
"""

combine_prompt_template = PromptTemplate(template=combine_prompt, input_variables=["text"])

llm = OpenAI(temperature=0.15)

chain = load_summarize_chain(llm, chain_type="map_reduce",map_prompt=map_prompt_template,combine_prompt=combine_prompt_template,verbose=True)
print(globalvar.inputText)
summary = chain.run(globalvar.inputText)
```

```
Rogue Score : {'rouge1': Score(precision=0.8011363636363636, recall=0.1206159110350727l, fmeasure=0.2096654275092936), 'rouge2': Score(precision=0.3428571428571286, recall=0.05136986301369863, fmeasure=0.0893521965748324l), 'rougeL
': Score(precision=0.4204545454545547, recall=0.06330196749358426, fmeasure=0.11003717472118961)}
```

12. Conclusion

The TSR project, harnessing Large Language Models, revolutionizes document understanding. Its recursive summarization and retrieval augmented generation ensure accurate, contextually rich insights. This pioneering approach has the potential to transform information retrieval across diverse domains. The system's applications span academia, content creation, business analysis, education, and general use, offering a powerful tool for efficient text processing. Overall, TSR represents a significant advancement in text summarization and insightful response generation, with far-reaching implications.

13. REFERENCES

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